

3.14). Give 3 possible ways to express the following Boolean Function with 8 or fewer literals.

$$F = A'B'C'D + AB'CD + A'B'C' + ACD' \quad (1010) \\ (0101) \quad (1011) + A'B'C'D' + A'B'C'D + AB'CD' \\ (0000) \quad (0001) + ABCD' \quad (1110).$$

AB \ CD	00	01	11	10
00	m ₀ 1	m ₁ 1	m ₃	m ₂
01	m ₄	m ₅ 1	m ₇	m ₆
11	m ₁₂	m ₁₃	m ₁₅	m ₁₄
10	m ₈	m ₉	m ₁₁ 1	m ₁₀ 1

$$F = A'B'C' + A'C'D + AB'C + ACD'$$

$$= A'C'(B'D) + AC(B'D') \\ \begin{matrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \end{matrix} \\ 8 \text{ literals} \rightarrow [1]$$

Consider $A'C'(B'D) + AC(B'D')$.

$$= B'(A'C' + AC) + DA'C' + D'AC.$$

$$= B'(A \oplus C) + DA'C' + D'AC \rightarrow (1)$$

Consider now $DA'C' + D'AC$.

$DA'C' + D'AC$ evaluates to 1 when:

- (i) $(A=C=0)$ OR $(A=C=1)$, which is $A \odot C$ AND when
- (ii) $(A'D=1)$ OR $(AD'=1)$, which is $A \oplus D$ OR when
- (iii) $(C'D=1)$ OR $(CD'=1)$, which is $C \oplus D$

$$\therefore (1) \text{ is } B'(A \odot C) + (A \odot C)[(A \oplus D) + (C \oplus D)]$$

But we also have $A=C$.

Hence $A \oplus D + C \oplus D$ is just *either* $A \oplus D$ *or* $C \oplus D$

$$\therefore (1) \text{ is } (A \odot C)[B' + A \oplus D] \text{ 5 literals.}$$

$$\text{OR } (A \odot C)[B' + C \oplus D] \text{ 5 literals.}$$

In general $XYZ' + X'Y'Z$ is $(X \odot Y)(X \oplus Z)$.

3.15b) $F(A, B, C, D) = \sum (0, 6, 8, 13, 14)$ $d(A, B, C, D) = \sum (2, 4, 10)$

AB \ CD	00	01	11	10
00	m ₀ 1	m ₁	m ₃	m ₂ X
01	m ₄ X	m ₅	m ₇	m ₆ 1
11	m ₁₂	m ₁₃ 1	m ₁₅	m ₁₄ 1
10	m ₈ 1	m ₉	m ₁₁	m ₁₀ X

$$= B'D' + A'D' + CD' + ABC'D$$

Is the Implication Operator associative.

Implication TT: $(x \rightarrow y)$

x	y	$x \rightarrow y$
0	0	1
0	1	1
1	0	0
1	1	1

y	z	$y \rightarrow z$
0	0	1
0	1	1
1	0	0
1	1	1

x	y	z	$y \rightarrow z$	$x \rightarrow (y \rightarrow z)$	$x \rightarrow y$	z	$(x \rightarrow y) \rightarrow z$
0	0	0	1	1	1	0	0
0	0	1	1	1	1	1	1
0	1	0	0	1	1	0	0
0	1	1	1	1	1	1	1
1	0	0	1	1	0	0	1
1	0	1	1	1	0	1	1
1	1	0	0	0	1	0	0
1	1	1	1	1	1	1	1

Hence $x \rightarrow (y \rightarrow z) \neq (x \rightarrow y) \rightarrow z$